



Scoring

ARC-AGI-3 Scoring Methodology

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How ARC-AGI-3 scoring works

ARC-AGI-3 uses **Relative Human Action Efficiency** (RHAЕ, pronounced "ray") to score AI systems.

RHAЕ measures per-level action efficiency compared to a human baseline, normalized per game, across all games.

What Gets Measured

AI is scored on two criteria:

1. **Completion** — How many levels did the AI complete in each game?
2. **Efficiency** — How many actions did the AI take compared to humans?

What Counts as an Action

An *action* is a discrete interaction with the environment. Each turn where the agent submits a command, move, or input that affects the game state counts as an action.

Internal operations that do not alter the environment (tool calls, reasoning steps, retries) are **not counted** as actions.

Human Baseline

Human baselines are established through controlled testing where participants play each ARC-AGI-3 game for the first time (having never seen the game before). For each game,



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upper of the two middle entries is selected. For example, if four players complete a level, third place is the baseline; if five players complete it, third place is still the baseline.

Using the upper median human per level:

- Reflects typical proficient human performance rather than outlier runs
- Reduces the impact of luck on any individual level
- Keeps the baseline grounded in real play, not theoretical speed-runs

How Scoring Works

Per-Level Scoring

For each level the AI completes, calculate:

$$\text{level_score} = (\text{human_baseline_actions} / \text{ai_actions}) ^ 2$$

- If human baseline is 10 actions and AI takes 10 → level score is 1.0 (100%)
- If human baseline is 10 actions and AI takes 20 → level score is 0.25 (25%)
- If human baseline is 10 actions and AI takes 1,00 → level score is 0.01 (1%)

Per-Level Score Cap

The maximum score per level is capped at **1.15x** human baseline. If an AI discovers a shortcut and completes a level faster than humans, it can receive at most 1.15.

This ensures a single subpar level does not disproportionately drag down the overall score for an AI that generalizes well across an entire game.

Per-Game Aggregation



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The maximum game score is also determined by this weighted average structure — it is capped based on how many levels the AI actually completed. To unlock a maximum game score of 100%, the AI must complete all levels, including the final one.

Example: A game has 5 levels and the AI completes only the first 4:

$$\text{max_game_score} = (1 + 2 + 3 + 4) / (1 + 2 + 3 + 4 + 5) = 10 / 15 = 66.7\%$$

No matter how efficiently the AI played levels 1–4, its game score cannot exceed 66.7%.

Total Score

Total score is the **average of all game scores**, resulting in a final score between 0% and 100%.

Score Interpretation

Score	Interpretation
100%	AI completes all games/levels while matching or surpassing human efficiency
1 - 99%	A mixture of level completion rates and efficiency relative to human baseline
0%	AI never completes a level across any game

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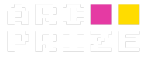
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